
Algorithm 1: Watermarking embedding, with repeated embedding.

```
1 /* Text  $c$  composed of  $n$  characters */
2  $T \leftarrow \{c_1, \dots, c_n\}$ ;
3 /* List of originals with one or two duplicates */
4  $OriginalsSingle \leftarrow \{0x516B, 0x66B4, \dots\}$ ;
5  $OriginalsDouble \leftarrow \{0x8FB0, 0x4E8C, \dots\}$ ;
6  $Punctuations \leftarrow \{0x002C, 0x003B, \dots\}$ ;
7 /* List of single and double duplicates */
8  $SingleDuplicates \leftarrow \{0x516B, 0x66B4, \dots\}$ ;
9  $DoubleDuplicates \leftarrow \{0x8FB0, 0x4E8C, \dots\}$ ;
10  $Spaces \leftarrow \{0x0020, 0x2000, \dots\}$ ;
11  $Confusables \leftarrow SingleDuplicates \cup DoubleDuplicates \cup Spaces$ ;
12  $WT \leftarrow []$ ;
13  $GetSingleDuplicate : OriginalsSingle \rightarrow SingleDuplicates$ ;
14  $GetDoubleDuplicate : (OriginalsDouble, \{0, 1\}) \rightarrow DoubleDuplicates$ ;
15  $GetSpace : \{000, \dots, 111\} \rightarrow Spaces$ ;
16  $i = 0$ ; /* Pointer to watermark index */
17 foreach  $c \in T$  do
18   if  $c \in Confusables$  then
19     if  $c \in Spaces$  then
20        $bits = []$ ;
21       for  $ii = 0..2$  do
22          $Append(bits, W[(i + ii)\%64])$ ;
23        $c \leftarrow GetSpace(bits)$ ;
24     else if  $c \in Punctuations$  then
25        $bits1 = []$ ;
26       for  $ii = 0..2$  do
27          $Append(bits1, W[(i + ii)\%64])$ ;
28        $bits2 = []$ ;
29       for  $ii = 0..2$  do
30          $Append(bits2, W[(i + ii)\%64])$ ;
31        $c \leftarrow GetSingleDuplicate(c) + GetSpaces(bits1) +$ 
32          $GetSpaces(bits2)$ ;
33     else
34        $bit = W[i\%64]$ ;
35       if  $bit = 1$  then
36         if  $c \in DoubleDuplicates$  then
37           if  $\neg Eof(W)$  then
38              $bit2 = W[i\%64]$ ;
39           else
40              $bit2 = W[0]$ ;
41            $c \leftarrow GetDoubleDuplicate(c, bit2)$ ;
42         else
43            $c \leftarrow GetSingleDuplicate(c)$ ;
44    $Append(WT, c)$ ;
45 return  $WT$ 
```

Algorithm 2: Watermark extraction.

```
1 /* Text string  $c$  composed of  $n$  characters */
2  $T \leftarrow \{c_1, \dots, c_n\}$ ;
3 /* List of originals with one or two duplicates */
4  $Originals \leftarrow \{0x516B, 0x66B4, \dots\}$ ;
5  $Punctuations \leftarrow \{0x002C, 0x003B, \dots\}$ ;
6 /* Lists of single and double duplicates */
7  $SingleDuplicates \leftarrow \{0x516B, 0x66B4, \dots\}$ ;
8  $DoubleDuplicates \leftarrow \{0x8FB0, 0x4E8C, \dots\}$ ;
9  $Spaces \leftarrow \{0x0020, 0x2000, \dots\}$ ;
10 /* Watermarking sequence */
11  $W \leftarrow []$ 
12 /* GetSpace inverse function */
13  $GetSpaceBit : Spaces \rightarrow \{000, \dots, 111\}$ 
14  $GetDoubleDupBit : DoubleDuplicates \rightarrow \{10, 11\}$ 
15 foreach  $c \in T$  do
16   if  $c \in Punctuations$  then
17     continue
18   else if  $c \in Spaces$  then
19      $Append(W, GetSpaceBit(c))$ ;
20   else if  $c \in DoubleDuplicates$  then
21      $Append(W, GetDoubleDupBit(c))$ ;
22   else if  $c \in SingleDuplicates$  then
23      $Append(W, 1)$ ;
24   else if  $c \in Originals$  then
25      $Append(W, 0)$ ;
26   if  $|W| \geq 64$  then
27     break
28 return  $W[: 64]$ 
```
